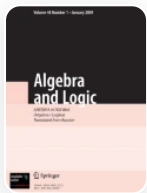


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Von Neumann Regular Hyperrings and Applications to Real Reduced Multirings

Published: 09 May 2024

Volume 62, pages 215–256, (2023) [Cite this article](#)

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Aims and scope

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A multiring is a ring-like structure where the sum is multivalued and a hyperring is a multiring with a strong distributive property. With every multiring we associate a structural presheaf, and when that presheaf is a sheaf, we say that the multiring is geometric. A characterization of geometric von Neumann hyperrings is presented. And we build a von Neumann regular hull for multirings which is used in applications to algebraic theory of quadratic forms. Namely, we describe the functor Q , introduced by M. Marshall in [J. Pure Appl. Alg., **205**, No. 2, 452–468 (2006)], as a left adjoint functor for the natural inclusion of the category of real reduced multirings (similar to real semigroups) into the category of preordered multirings and explore some of its properties. Next, we employ sheaf-theoretic methods to characterize real reduced hyperrings as certain geometric von Neumann regular real hyperrings and construct the functor V , a geometric von Neumann regular hull for a multiring. Finally, we look at some interesting logical and algebraic interactions between the functors Q and V that are useful for describing hyperrings in the

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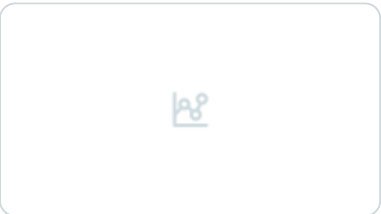
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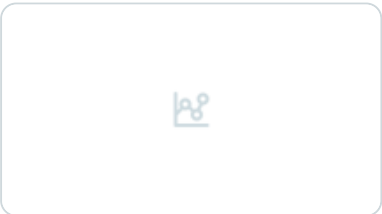
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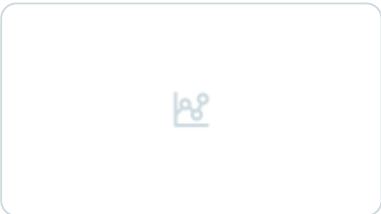
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Acknowledgements

We express our sincere gratitude to the referee for his/her careful reading and valuable suggestions that significantly improved the presentation of the text.

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Additional information

Translated from *Algebra i Logika*, Vol. 62, No. 3, pp. 323–386, May–June, 2023. Russian DOI: <https://doi.org/10.33048/alglog.2023.62.302>.

Supported by Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq). (H. R. O. Ribeiro)

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Cite this article

Ribeiro, H.R.O., Mariano, H.L. Von Neumann Regular Hyperrings and Applications to Real Reduced Multirings. *Algebra Logic* **62**, 215–256 (2023). <https://doi.org/10.1007/s10469-024-09739-0>

Received

12 November 2022

Accepted

10 April 2024

Published

09 May 2024

Issue Date

July 2023

DOI

<https://doi.org/10.1007/s10469-024-09739-0>

Keywords

[real algebra](#)

[quadratic form](#)

[multiring](#)

[von Neumann hyperring](#)

[real reduced multiring](#)

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